

Exercise 3i

- ① Do the following additions in the given bases.
 - a $1011 + 1011$ (base two)
 - b $421 + 12 + 42$ (base five)
 - c $774 + 447 + 113$ (base eight)
 - d $t0e + tee$ (base twelve)
- ② Do the following subtractions in the given bases.
 - a $1101 - 11$ (base two)
 - b $321 - 113$ (base five)
 - c $405 - 217$ (base eight)
 - d $905 - 509$ (base twelve)
- ③ Simplify the following. Express each answer in the given base.
 - a $402_{five} \times 13_{five}$
 - b $341_{five} \div 22_{five}$
 - c $48_{twelve} \times 15_{twelve}$
 - d $356_{eight} \div 7_{eight}$
- ④ All the numbers below are binary. Evaluate
 - a 1101×11
 - b 10011×101
 - c 11101×111
 - d 101011×110
 - e $100100 \div 11$
 - f $100011 \div 101$
 - g $101101 \div 101$
 - h $111100 \div 110$
- ⑤ Find the missing number (****) if the following addition is in base five.

$$\begin{array}{r}
 2004 \\
 143 \\
 **** \\
 + 201 \\
 \hline
 4240
 \end{array}$$

3.5 The binary system (revision)

Binary numbers

Here we **define** base two numbers as belonging to the **binary system**. So, for example 11010_{two} and 111_{two} are **binary numbers**. The binary system is important because of its relationship to computer circuits. Binary numbers only have two digits: 0 and 1. Likewise a computer circuit can only be in two states, either off (0) or on (1).

Table 3.1 shows the first ten binary numbers.

binary number	value in powers of 2	base ten number
1	1×1	1
10	$1 \times 2^1 + 0 \times 1$	2
11	$1 \times 2^1 + 1 \times 1$	3
100	$1 \times 2^2 + 0 \times 2^1 + 0 \times 1$	4
101	$1 \times 2^2 + 0 \times 2^1 + 1 \times 1$	5
110	$1 \times 2^2 + 1 \times 2^1 + 0 \times 1$	6
111	$1 \times 2^2 + 1 \times 2^1 + 1 \times 1$	7
1000	$1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 0 \times 1$	8
1001	$1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 1$	9
1010	$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 1$	10

Table 3.1

To add, subtract and multiply binary numbers, use the methods that you use with base ten numbers. However, you must remember that you are working with powers of two, not powers of ten. The following identities are very useful:

Addition:

$$\begin{array}{ll}
 0 + 0 = 0 & 1 + 0 = 1 \\
 0 + 1 = 1 & 1 + 1 = 10
 \end{array}$$

Multiplication:

$$\begin{array}{ll}
 0 \times 0 = 0 & 1 \times 0 = 0 \\
 0 \times 1 = 0 & 1 \times 1 = 1
 \end{array}$$

Exercise 3j

- ① Figure 3.1 shows how to use a piece of paper and three bottle tops to represent the binary number 1011_{two} .

2^3	2^2	2^1	2^0
			

Draw a piece of paper like this and collect some counters (bottle tops, pebbles).

- a Using one counter, represent as many different binary numbers as possible on the paper.
- b Repeat with two counters.
- c Repeat with three counters.
- d Repeat with four counters.
- e Arrange your binary numbers in order of size.

- 2 Table 3.1 contains the first ten binary numbers. Extend the table to include the first twenty binary numbers.

- 3 a Convert 10_{ten} to a binary number.
b Convert $(10_{ten})^2$ to a binary number.

- 4 Use repeated division to convert the following denary (i.e. base ten) to binary numbers.

a 28 b 27 c 38 d 41
e 50 f 59 g 63 h 77

- 5 Convert the following binary numbers to base ten numbers.

a 1001 b 1011 c 10101
d 11110 e 11010 f 110101

- 6 Convert 111110_{two} to
a a base ten number
b a base eight number

- 7 Find the value of the following, leaving your answers in base two.

a the square of 111_{two}
b $(1011_{two})^2$

- 8 Find the square root of the following. Give your answers in base two.

a 1000000_{two}
b 1010001_{two}

- 9 Calculate the following in base two, checking your results by converting to base ten.

a $\begin{array}{r} 1101 \\ + 1011 \\ \hline \end{array}$ b $\begin{array}{r} 10110 \\ + 111 \\ \hline \end{array}$

c $\begin{array}{r} 11101 \\ - 1011 \\ \hline \end{array}$ d $\begin{array}{r} 11110 \\ - 111 \\ \hline \end{array}$

e $\begin{array}{r} 1101101 \\ + 11011 \\ \hline \end{array}$ f $\begin{array}{r} 10100 \\ - 1110 \\ \hline \end{array}$

g $\begin{array}{r} 1100 \\ \times 11 \\ \hline \end{array}$ h $\begin{array}{r} 10011 \\ \times 101 \\ \hline \end{array}$

i $\begin{array}{r} 1111 \\ \times 110 \\ \hline \end{array}$ j $\begin{array}{r} 10111 \\ \times 1011 \\ \hline \end{array}$

- 10 Calculate:

a $111_{two} \times (111_{two} + 1010_{two})$
b $111_{two} \times (1111_{two} - 1010_{two})$

- 11 Use long multiplication to check that:
 $(11_{two})^4 = (1001_{two})^2 = 1010001_{two}$.

Bicimals

Base ten fractions, or decimals, are based on negative powers of ten (Fig. 3.2).

$$\begin{array}{ccccccc} 6 \times 10^0 & 5 \times 10^{-1} & 8 \times 10^{-2} & 3 \times 10^{-3} \\ & \swarrow & \uparrow & \searrow \\ & 6.583 & & \end{array}$$

Fig. 3.2

Similarly we can have base two fractions, **bicimals**, based on negative powers of two (Fig. 3.3).

$$\begin{array}{ccccccc} 1 \times 2^0 & 1 \times 2^{-1} & 0 \times 2^{-2} & 1 \times 2^{-3} \\ & \swarrow & \uparrow & \searrow \\ & 1.101 & & \end{array}$$

Fig. 3.3

To convert a bicimal to a decimal, first express each digit as a power of two, then change the powers to fractions. Read Example 20 carefully.

Example 20

Convert the following bicimals to decimals.

a 1.101 b 10.011 c 110.11

$$\begin{aligned} \text{a } 1.101 &= 1 \times 2^0 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ &= 1 + 1 \times \frac{1}{2} + 0 \times \frac{1}{2^2} + 1 \times \frac{1}{2^3} \\ &= 1 + \frac{1}{2} + 0 + \frac{1}{8} \\ &= 1 + 0.5 + 0 + 0.125 \\ &= 1.625 \end{aligned}$$

$$\begin{aligned} \text{b } 10.011 &= 1 \times 2^1 + 0 \times 2^0 + 0 \times 2^{-1} \\ &\quad + 1 \times 2^{-2} + 1 \times 2^{-3} \\ &= 2 + 0 + 0 \times \frac{1}{2} + 1 \times \frac{1}{2^2} + 1 \times \frac{1}{2^3} \\ &= 2 + 0 + 0 + \frac{1}{4} + \frac{1}{8} \\ &= 2 + 0.25 + 0.125 \\ &= 2.375 \end{aligned}$$

$$\begin{aligned}
 c \quad 110.11 &= 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\
 &\quad + 1 \times 2^{-1} + 1 \times 2^{-2} \\
 &= 4 + 2 + 0 + \frac{1}{2} + \frac{1}{4} \\
 &= 6 + 0.5 + 0.25 \\
 &= 6.75
 \end{aligned}$$

Example 21

Calculate:

a $10.101 + 1.101$, **b** $11.01 - 1.11$.

The working is shown without explanation:

$$\begin{array}{r}
 \text{a} \quad 10.001 \\
 + 1.101 \\
 \hline
 100.010
 \end{array}
 \qquad
 \begin{array}{r}
 \text{b} \quad 11.01 \\
 - 1.11 \\
 \hline
 1.10
 \end{array}$$

Exercise 3k

- ① Copy and complete Table 3.2.

Some powers of two	2^{-1}	2^{-2}	2^{-3}	2^{-4}
As common fractions	$\frac{1}{2}$	$\frac{1}{4}$		
As decimal fractions	0.5			
As bicimals	0.1	0.01		

Table 3.2 Some negative powers of two

- ② Convert the following bicimals to base ten.

a 0.1 **b** 0.01 **c** 0.001
d 0.11 **e** 0.101 **f** 0.111
g 11.1 **h** 10.01 **i** 10.0001

- ③ Simplify each of the following. Give your answers **i** in base two, **ii** in base ten.

a $10.1 + 1.11$ **b** $1.101 + 0.001$
c $11.11 - 1.01$ **d** $10.01 - 1.1$

- ④ Calculate

a $0.101 + 0.011$ **b** $101.01 + 1.01$
c $11.001 - 0.101$ **d** $10.1 - 1.01$

SUMMARY

- 1 When adding and subtracting fractions, express the given fractions as equivalent fractions with the same denominators (i.e. common denominators).
- 2 When adding or subtracting decimals, always arrange the numbers so that decimal points are under each other.
- 3 $x\%$ is a short way of writing $\frac{x}{100}$ or 'x per cent'. When finding a percentage increase or percentage decrease, always calculate the percentage on the original amount.

- 4 Simple interest, I , is interest paid at regular intervals of time. It is calculated by the formula

$$I = \frac{PRT}{100}$$

where P is the principal, R is the rate per cent per annum, and T is the corresponding time in years.

- 5 Numbers can be expressed in any base from base two (*binary*) upwards. However, for bases beyond base ten (*denary*), it is necessary to invent new symbols.
- 6 To convert a number from a given base to base ten, either express the number as powers of the base and add, or use repeated multiplication. (See Example 17.)
- 7 To convert a number from base ten to base n , divide the number repeatedly by n until further division is impossible. Then read the remainders upwards. (See Example 16b.)
- 8 It is possible to add, subtract, multiply and divide in any base. (See Examples 18 and 19.) Note, however, that calculations are normally in base ten.
- 9 A *bicimal* is a base two fraction (equivalent to decimal in base ten).

4

Algebraic processes (1) Simplification and substitution

OBJECTIVES

By the end of this chapter you should be able to:

- 1 Use letters of the alphabet to represent numbers
- 2 Simplify algebraic expressions by grouping like terms, by removing brackets, by expanding brackets
- 3 Add and subtract algebraic fractions
- 4 Factorise algebraic expressions by determining common factors and by grouping terms
- 5 Substitute values into given algebraic expressions.

Teaching and learning materials

Teacher: Factorisation charts.

4.1 Use of letters

Algebra uses letters of the alphabet to represent general numbers. The easiest way of dealing with letters that represent numbers is to imagine that a letter stands for a particular number.

Example 1

Express p metres in centimetres.

Since $1 \text{ m} = 100 \text{ cm}$
 $2 \text{ m} = 100 \times 2 \text{ cm}$
 $3 \text{ m} = 100 \times 3 \text{ cm}$
 Hence $p \text{ m} = 100 \times p \text{ cm}$
 $= 100p \text{ cm}$

In Example 1, notice that $100p$ is short for $100 \times p$. In the expression $100p$, the 100 is called the **coefficient** of p .

Example 2

In a class of p students the average mark is x and in another class of n students the average mark is y . What is the average mark for both classes? [WAEC]

$$\text{Average mark} = \frac{\text{total marks scored}}{\text{number of students}}$$

$$\begin{aligned} \text{In 1st class, total marks scored} &= p \times x \\ &= px \end{aligned}$$

$$\begin{aligned} \text{In 2nd class, total marks scored} &= n \times y \\ &= ny \end{aligned}$$

$$\text{For both classes, total marks scored} = px + ny$$

$$\text{Number of students} = p + n$$

$$\text{Average mark} = \frac{px + ny}{p + n}$$

Exercise 4a

- 1 A student has 200 naira. If someone gives her 160 naira, how much will she have? If someone gives her x naira how much will she have?
- 2 A student is 15 years old. How old will he be in 4 years' time? How old will he be in y years' time?
- 3 A woman has ₦2 000. If she spends ₦800, how much will she have left? If she spends ₦ a instead, how much will she have left?
- 4 A man was 20 years old 6 years ago. How old is he now? p years ago another man was 23 years old. How old is he now?
- 5 A lorry carries 8 t of blocks.
 - a If it delivers 3 t, what mass of blocks is left on the lorry?
 - b If it delivers d t, what mass remains on the lorry?
- 6 What number is
 - a five times as big as 20,
 - b five times as big as x ?

- 7 How many centimetres in
a 3 metres b e metres?
- 8 How many hours in
a 120 min b m min?
- 9 A sheet of paper has an area of 300 cm^2 .
a How long is the paper if its breadth is 15 cm?
b How long is the paper if its breadth is b cm?
- 10 Express x metres and y cm in cm.
- 11 Express a metres and b cm in cm.
- 12 The perimeter of a rectangle is 14 m and the length is x m. Express the breadth of the rectangle in terms of x .
- 13 The perimeter of a rectangle is 20 m and the length is x m. Find the area of the rectangle in terms of x . [WAEC]
- 14 A rectangle has one side equal to $(2x + 4)$ cm and a perimeter of $(6x + 4)$ cm. What is the area of the rectangle in terms of x ? [WAEC]
- 15 A car travels d km at an average speed of u km/h. How long does it take?
- 16 A train moves at an average speed of x km/h. How many hours will it take to cover 340 km? [WAEC]
- 17 A train travels for t hours at a speed of v km/h. How far does it go?
- 18 Two cars start together from the same place and travel in the same direction along a road. If the faster one has an average speed of u km/h and the other v km/h, how far ahead is the faster one after t hours? [WAEC]
- 19 A boy's age is x years and his father is four times as old. Find the father's age in y years' time.
- 20 p years ago a man's age was $(q + r)$ times that of his son who was then r years old. How old is the man now? [WAEC]

4.2 Simplification

Grouping terms

Expressions such as $2x$, x , $\frac{1}{3}x$ and $-5x$ are called **terms in x** . Since they are *all* terms in x , we say they are **like terms**. Expressions containing a number of like terms can be simplified by *grouping* them together.

Example 3

Simplify $9a - 5a - 8a + 10a$.

Either, by grouping positive and negative terms together,

$$\begin{aligned} 9a - 5a - 8a + 10a \\ &= 9a + 10a - 5a - 8a \\ &= 19a - 13a \\ &= 6a \end{aligned}$$

or, by treating the terms as directed numbers,

$$\begin{aligned} 9a - 5a - 8a + 10a \\ &= 4a - 8a + 10a \\ &= -4a + 10a \\ &= 6a \end{aligned}$$

Example 4

Simplify $5x - 8x + x + 3y$.

$$\begin{aligned} 5x - 8x + x + 3y \\ &= -3x + x + 3y \\ &= -2x + 3y \\ &= 3y - 2x \end{aligned}$$

In Example 4, notice that $2x$ and $3y$ are **unlike terms**. $3y - 2x$ cannot be simplified any further.

Exercise 4b

Simplify the following expressions as far as possible.

- | | |
|------------------|------------------|
| 1 $9x - 2x$ | 2 $2a - 9a$ |
| 3 $-3y - 30y$ | 4 $2y + 5y - 3y$ |
| 5 $4x - 8x + 9x$ | 6 $9z - 3z - z$ |
| 7 $3a - 7a - 2a$ | 8 $b + 4b - 12b$ |

- 9 $6c - 17c + 3c$
- 10 $2a + 5x - 3a$
- 11 $-3h - 6g + 10g$
- 12 $8d - 3 - 7d$
- 13 $3a - 5a + 11a - 4a$
- 14 $7x + 3x - x - 5x$
- 15 $8k - 4k + 3k - 7k$
- 16 $6x - 9x + 2x + 4y$
- 17 $2a - 3b + 5b - 8a$
- 18 $3x + 8y - 5x - y$
- 19 $2m - 9n - 5m + 4n$
- 20 $r - 3s - 3t - 4r + 10s + 8t$

Removing brackets

If any quantity multiplies the terms inside a bracket, every term inside the bracket must be multiplied by that quantity when the bracket is removed.

In general, $a(x + y) = ax + ay$
and $a(x - y) = ax - ay$

Example 5

Remove the brackets from $4(3x - 5y + z)$.

$$\begin{aligned} 4(3x - 5y + z) \\ &= 4 \times 3x - 4 \times 5y + 4 \times z \\ &= 12x - 20y + 4z \end{aligned}$$

Example 6

Remove brackets and simplify $3 - (a - \overline{5 - 6a})$.

Note: the line over the terms $5 - 6a$ acts like a bracket.

$$\begin{aligned} 3 - (a - \overline{5 - 6a}) \\ &= 3 - (a - 5 + 6a) \\ &= 3 - (7a - 5) \\ &= 3 - 7a + 5 \\ &= 8 - 7a \end{aligned}$$

In Example 6, notice that we remove the inner bracket *first*. Notice also that a negative sign outside a bracket is equivalent to -1 outside the bracket, i.e. $-(5 - 6a) = -1(5 - 6a)$.

Example 7

Remove brackets and simplify

$$3x(2 - x) - 5(6 + x - 2x^2).$$

$$\begin{aligned} 3x(2 - x) - 5(6 + x - 2x^2) \\ &= 6x - 3x^2 - 30 - 5x + 10x^2 \\ &= 10x^2 - 3x^2 + 6x - 5x - 30 \\ &= 7x^2 + x - 30 \end{aligned}$$

Example 8

Simplify **a** $3 \times 2a - 5$, **b** $3(2a - 5)$.

$$\mathbf{a} \quad 3 \times 2a - 5 = 6a - 5$$

$$\mathbf{b} \quad 3(2a - 5) = 3 \times 2a - 3 \times 5 \\ = 6a - 15$$

Exercise 4c

Remove brackets from the following and simplify where possible.

- 1 $3(a + b - 2)$
- 2 $-2(3m - n + 4)$
- 3 $2a + 3(a + 3b)$
- 4 $5(2y - x) + 6x$
- 5 $\frac{1}{2}(2u - 8) + 5$
- 6 $3c - \frac{2}{3}(12c - 3d)$
- 7 $4 \times 3m - 2$
- 8 $4 \times (3m - 2)$
- 9 $4u - \overline{7u - 3v}$
- 10 $5d - \overline{e + 2d}$
- 11 $2(a - 3b) + 3(a + b)$
- 12 $5m + 5(2n - m) - 8n$
- 13 $2(h + 5k) + 5(h + 2k)$
- 14 $5(2u - 3v + 4w) - 7(u - 2v + 3w)$
- 15 $4(r - 3s - 3t) - 2(2r - 5s - 4t)$ [WAEC]
- 16 $6x - (3 - \overline{x - 8})$
- 17 $2a - (4a - \overline{5a - 7})$ [WAEC]
- 18 $2x(3x - 1) + 7(x^2 + 3)$
- 19 $6(a^2 - 2a - 1) - 3a(2a - 5)$
- 20 $4x(3x - 2y) + 5y(x + 3y) - 7xy$

Expansion of brackets

The expression $(a + b)(c + d)$ means $(a + b) \times (c + d)$. The terms in the first bracket multiply each term in the second bracket.

$$\begin{aligned}(a + b)(c + d) \\ &= (a + b)c + (a + b)d \\ &= ac + bc + ad + bd\end{aligned}$$

Alternatively, each term in the first bracket multiplies the terms in the second bracket.

$$\begin{aligned}(a + b)(c + d) \\ &= a(c + d) + b(c + d) \\ &= ac + ad + bc + bd \\ &= ac + bc + ad + bd\end{aligned}$$

In either case, $ac + bc + ad + bd$ is the **expansion** of $(a + b)(c + d)$.

Example 9

Expand the following, simplifying where possible.

a $(r - x)(r + y)$

b $(x - 5)(x + 8)$

c $(2a + 3b)(5a - 2b)$

d $(m + 4n)^2$

a $(r - x)(r + y) = r(r - x) + y(r - x)$
 $= r^2 - rx + ry - xy$

b $(x - 5)(x + 8) = x(x - 5) + 8(x - 5)$
 $= x^2 - 5x + 8x - 40$
 $= x^2 + 3x - 40$

c $(2a + 3b)(5a - 2b)$
 $= 5a(2a + 3b) - 2b(2a + 3b)$
 $= 10a^2 + 15ab - 4ab - 6b^2$
 $= 10a^2 + 11ab - 6b^2$

d $(m + 4n)^2 = (m + 4n)(m + 4n)$
 $= m(m + 4n) + 4n(m + 4n)$
 $= m^2 + 4mn + 4mn + 16n^2$
 $= m^2 + 8mn + 16n^2$

In Example 9d, the expression $(m + 4n)^2$ is called a **perfect square**. The expansions in parts **b**, **c** and **d** of Example 9 were further simplified by collecting the like terms together.

Exercise 4d

Expand the following, collecting like terms where possible.

- | | |
|------------------------|----------------------|
| ① $(p + q)(r + s)$ | ② $(6 + y)(5 + x)$ |
| ③ $(a - b)(c - d)$ | ④ $(2x + 1)(3x - 1)$ |
| ⑤ $(2n + 3)(2n + 5)$ | ⑥ $(4m - 9)(2m - 3)$ |
| ⑦ $(5x - 2)(x + 4)$ | ⑧ $(a + 5)(3a - 2)$ |
| ⑨ $(x - 8)(x + 9)$ | ⑩ $(3h + 4)(5h - 2)$ |
| ⑪ $(b - 5)(b + 5)$ | ⑫ $(c + 7)(c - 7)$ |
| ⑬ $(3t - 2)^2$ | ⑭ $(2x - 5y)^2$ |
| ⑮ $(5x - y)(2x - 3y)$ | ⑯ $(4a - 1)(5a - 2)$ |
| ⑰ $(5a + 2d)(3a - 8d)$ | ⑱ $(2d + e)(3d + e)$ |
| ⑲ $(6m + n)(5m - n)$ | ⑳ $(2k - 7)(4k + 5)$ |

Example 10

Find the coefficient of n in $(4n + 3)(n - 5) - (2n - 3)^2$.

Either

$$\begin{aligned}(4n + 3)(n - 5) - (2n - 3)^2 \\ &= n(4n + 3) - 5(4n + 3) - [2n(2n - 3) - 3(2n - 3)] \\ &= 4n^2 + 3n - 20n - 15 - (4n^2 - 6n - 6n + 9) \\ &= 4n^2 - 17n - 15 - (4n^2 - 12n + 9) \\ &= 4n^2 - 17n - 15 - 4n^2 + 12n - 9 \\ &= -5n - 24\end{aligned}$$

The coefficient of n is -5 .

Or, by inspection:

For $(4n + 3)(n - 5)$, the term in n is found by adding the products of the inner and outer pairs of terms.

$$(4n + 3)(n - 5)$$

$$+3n + (-20n) = -17n$$

$(2n - 3)^2 = (2n - 3)(2n - 3)$. Again, the term in n is the sum of the products of the inner and outer terms of the brackets.

$$(2n - 3)(2n - 3)$$

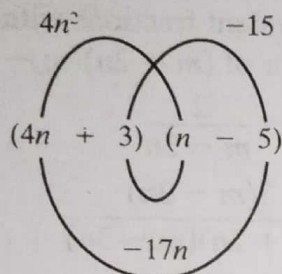
$$-6n + (-6n) = -12n$$

For the whole expression,

$$\begin{aligned}\text{term in } n &= -17n - (-12n) \\ &= -17n + 12n \\ &= -5n\end{aligned}$$

Coefficient of n is -5 .

Fig. 4.1 shows how to expand brackets directly by multiplying the terms inside the brackets in pairs.



$$(4n + 3)(n - 5) = 4n^2 - 17n - 15$$

Fig. 4.1

Some students remember this by saying, 'eyebrow, eyebrow, nose and mouth'.

Exercise 4e

- ① Either by expanding the brackets or otherwise, write down the coefficient of x in the following.
 - a $(x + 7)(x + 9)$
 - b $(x - 3)(x + 8)$
 - c $(x + 4)(x - 17)$
 - d $(2x + 5)(4x + 7)$
 - e $(3x - 1)(5x + 2)$
 - f $(2x - 8)(2x + 8)$
- ② Multiply $(3 + a)$ by $(5 - 2a)$. [WAEC]
- ③ Simplify the following.
 - a $(a - 3)(a - 4) + (a - 1)(a + 5)$
 - b $(m + 6)(m - 2) - (m - 3)(m + 1)$
 - c $(2x + 3)(x - 2) + 2x(x - 1)$
 - d $3d(2d + 3) - (3d + 1)(2d + 1)$
- ④ Simplify $(f + g)^2 + (f - 2g)^2$.
- ⑤ Simplify $(3b - 1)^2 - (2b + 1)^2$.
- ⑥ Simplify $(2a - 3b)^2 - (a - b)^2$. [WAEC]
- ⑦ Simplify $(y + z)^2 - 2yz$. [WAEC]

- ⑧ Find the coefficient of
 - a x^2 ,
 - b x in $(x + 1)(2x - 3) + (x - 4)^2$.
- ⑨ Simplify $(a + b)^2 - (a - b)^2$.
- ⑩ Simplify $(3n + t)(n + 3t) - 3(n - 2t)(n - 4t)$.

4.3 Adding and subtracting algebraic fractions

Just as in arithmetic, algebraic fractions must have a common denominator before they can be added or subtracted.

Example 11

Simplify the following.

$$\begin{array}{ll}\text{a } \frac{3x}{7} + \frac{2x}{7} & \text{b } \frac{5}{2a} - \frac{3}{2a} \\ \text{c } \frac{1}{4x} + \frac{2}{3x} & \text{d } \frac{7}{3a} - \frac{6}{5c}\end{array}$$

$$\text{a } \frac{3x}{7} + \frac{2x}{7} = \frac{3x + 2x}{7} = \frac{5x}{7}$$

$$\text{b } \frac{5}{2a} - \frac{3}{2a} = \frac{5 - 3}{2a} = \frac{2}{2a} = \frac{2 \div 2}{2a \div 2} = \frac{1}{a}$$

c The LCM of $4x$ and $3x$ is $12x$. $12x$ is the common denominator.

$$\begin{aligned}\frac{1}{4x} + \frac{2}{3x} &= \frac{1 \times 3}{4x \times 3} + \frac{2 \times 4}{3x \times 4} \\ &= \frac{3}{12x} + \frac{8}{12x} \\ &= \frac{3 + 8}{12x} \\ &= \frac{11}{12x}\end{aligned}$$

d The LCM of $3a$ and $5c$ is $15ac$. $15ac$ is the common denominator.

$$\begin{aligned}\frac{7}{3a} - \frac{6}{5c} &= \frac{7 \times 5c}{3a \times 5c} - \frac{6 \times 3a}{5c \times 3a} \\ &= \frac{35c}{15ac} - \frac{18a}{15ac} \\ &= \frac{35c - 18a}{15ac}\end{aligned}$$

This does not simplify further.

$\frac{x-2}{4}$ is a short way of writing $\frac{(x-2)}{4}$ or $\frac{1}{4}(x-2)$. Similarly $\frac{3x}{x-y}$ is a short way of writing $\frac{3x}{(x-y)}$. In each case, consider the terms inside the brackets to be a single term until the brackets can be properly removed.

Example 12

Simplify the following.

a $\frac{4a+13}{5} - \frac{2a+3}{3}$

b $2 + \frac{5(2a+1)}{6a} - \frac{4b-3}{2b}$

a The LCM of 5 and 3 is 15. 15 is the common denominator.

$$\begin{aligned} \frac{4a+13}{5} - \frac{2a+3}{3} &= \frac{3(4a+13)}{5 \times 3} - \frac{5(2a+3)}{3 \times 5} \\ &= \frac{3(4a+13) - 5(2a+3)}{15} \\ &= \frac{12a+39-10a-15}{15} \\ &= \frac{2a+24}{15} \end{aligned}$$

b The LCM of $6a$ and $2b$ is $6ab$. Make equivalent fractions with denominators of $6ab$.

$$\begin{aligned} 2 + \frac{5(2a+1)}{6a} - \frac{4b-3}{2b} &= \frac{6ab \times 2}{6ab} + \frac{b \times 5(2a+1)}{6ab} \\ &\quad - \frac{3a \times (4b-3)}{6ab} \\ &= \frac{12ab + 5b(2a+1) - 3a(4b-3)}{6ab} \\ &= \frac{12ab + 10ab + 5b - 12ab + 9a}{6ab} \\ &= \frac{10ab + 9a + 5b}{6ab} \end{aligned}$$

This does not simplify further.

Example 13

Express each of the following as a single fraction in its simplest form.

a $\frac{3}{m+2n} - \frac{2}{m-3n}$ **b** $\frac{x}{x+2} - \frac{x-2}{x-3}$

a The denominators are $(m+2n)$ and $(m-3n)$. Their LCM is $(m+2n)(m-3n)$. Make equivalent fractions with a common denominator of $(m+2n)(m-3n)$.

$$\begin{aligned} \frac{3}{m+2n} - \frac{2}{m-3n} &= \frac{3(m-3n)}{(m+2n)(m-3n)} \\ &\quad - \frac{2(m+2n)}{(m+2n)(m-3n)} \\ &= \frac{3(m-3n) - 2(m+2n)}{(m+2n)(m-3n)} \\ &= \frac{3m-9n-2m-4n}{(m+2n)(m-3n)} \\ &= \frac{m-13n}{(m+2n)(m-3n)} \end{aligned}$$

This does not simplify further.

b $\frac{x}{x+2} - \frac{x-2}{x-3}$

$$\begin{aligned} &= \frac{x(x-3) - (x+2)(x-2)}{(x+2)(x-3)} \\ &= \frac{x^2 - 3x - (x^2 - 4)}{(x+2)(x-3)} \\ &= \frac{x^2 - 3x - x^2 + 4}{(x+2)(x-3)} \\ &= \frac{4-3x}{(x+2)(x-3)} \end{aligned}$$

Exercise 4f

① Express each of the following as a single fraction.

a $\frac{4x}{9} + \frac{x}{9}$

b $\frac{5y}{2} - \frac{2y}{3}$

c $\frac{3}{a} - \frac{2}{b}$

d $\frac{1}{2x} + \frac{1}{7x}$

e $\frac{3}{4a} - \frac{3}{4b}$

f $\frac{3}{2ab} + \frac{4}{3bc}$

② Express $\frac{2}{a} + \frac{7}{b} - \frac{3}{c}$ as a single fraction. [WAEC]

③ Reduce $\frac{3}{x} - \frac{x}{2} + 5$ to a single fraction. [WAEC]

④ Simplify $\frac{5}{2cd} + \frac{4}{3de}$. [WAEC]

⑤ Simplify each of the following.

a $\frac{2x-3}{3} + \frac{x-1}{5}$

b $\frac{7x+2}{5} - \frac{5x+3}{5}$

c $\frac{a-2}{3} - \frac{a-4}{5}$

d $\frac{1}{2}(3x+1) + \frac{1}{3}(x+2)$

e $\frac{2(3x-2)}{5} + \frac{x+5}{6}$

f $\frac{3a-4}{5} - \frac{2a+19}{15}$

⑥ Simplify the following expression.

$\frac{3x+2}{3} - \frac{x-1}{4} - \frac{5}{12}$ [WAEC]

⑦ Simplify the following.

a $\frac{6x+1}{2a} - \frac{x-2}{2a}$

b $\frac{x-2}{6a} + \frac{2x+1}{3x}$

c $\frac{4a+1}{a} + \frac{3b-2}{b}$

d $3 - \frac{6a-5}{2a}$

e $\frac{2a-1}{5a} - \frac{4b-3}{10b}$

f $\frac{2x+1}{x} + \frac{3y-2}{y} - 5$

⑧ Express $\frac{a-4}{2b} - \frac{b-3}{6b} + 4$ as a single fraction in its lowest terms. [WAEC]

⑨ Simplify the following.

a $3 + \frac{2b}{a-b}$

b $2 - \frac{x}{x+2y}$

c $\frac{5}{a+4} - \frac{2}{a-2}$

d $\frac{2}{t+1} + \frac{3}{t+2}$

e $\frac{3x}{x-1} - \frac{x}{x+2}$

f $\frac{x}{x-3} - \frac{8}{x} - 2$

g $\frac{1}{n-6} + \frac{1}{n-4} - \frac{2}{n-5}$

h $\frac{x-2}{x+2} - \frac{x-1}{x+3}$

10 If $X = \frac{2a+3}{3a-2}$, express $\frac{X-1}{2X+1}$ in terms of a . [WAEC]

4.4 Factorisation

By common factor

Example 14

Complete the bracket in the statement

$8p - 20q = 4(\quad)$.

4 is the HCF of $8p$ and $20q$. Divide $8p$ and $20q$ by 4 to find the terms inside the bracket.

$8p \div 4 = 2p$ and $20q \div 4 = 5q$

$8p - 20q = 4(2p - 5q)$

In Example 14, 4 and $(2p - 5q)$ are **factors** of $8p - 20q$. 4 is the **highest common factor** of the given terms.

Example 15

Factorise the following.

a $2a^3 - 5a^2 - a$

b $15a^3b^4c - 6a^2b^5c^2$

a The common factor is a .

$2a^3 - 5a^2 - a = a(2a^2 - 5a - 1)$

b The HCF of the two terms is $3a^2b^4c$.

$15a^3b^4c - 6a^2b^5c^2 = 3a^2b^4c(5a - 2bc)$

Check the results of Example 15 by multiplication.

Exercise 4g (Oral)

Factorise the following either by completing the brackets or by finding the highest common factors of the given terms.

① $2m + 8n = 2(\quad)$

② $3a - 15b = 3(\quad)$

③ $10x - 5 = 5(\quad)$

④ $-3h - 12k = -3(\quad)$

⑤ $-2x + 18 = -2(\quad)$

- ⑥ $5a - 8ab = a(\quad)$
- ⑦ $9x + 3xz = 3x(\quad)$
- ⑧ $8cm + 12dm - 16em = 4m(\quad)$
- ⑨ $3x^3 - 12x^2 - 9x = 3x(\quad)$
- ⑩ $10a^2b^2 - 15a^2b + 20ab^2 = 5ab(\quad)$
- ⑪ $4a - 8b$
- ⑫ $9x + 12y$
- ⑬ $3ab - 6ac + 3ad$
- ⑭ $8px - 4qx + 8rx$
- ⑮ $3m^3 - 2m^2 + m$
- ⑯ $6n^4 - 2n^3 + 4n^2$
- ⑰ $5ab + 4a^2b - 6ab^2$
- ⑱ $2abx + 7acx - 3a^2x$
- ⑲ $4a^4 + 2a^3b - 10a^2b^2$
- ⑳ $24a^2bc^2 + 30a^2c^2x - 18a^3cx^2$

Example 16

Factorise $12x^2 + 3x - 4x - 1$.

The terms $12x^2$ and $3x$ have $3x$ in common.
The terms $-4x$ and -1 have -1 in common.

$$\begin{aligned} 12x^2 + 3x - 4x - 1 &= 3x(4x + 1) - 1(4x + 1) \\ &= (4x + 1)(3x - 1) \text{ since the expressions } \\ &\quad 3x(4x + 1) \text{ and } -1(4x + 1) \text{ both have } \\ &\quad (4x + 1) \text{ in common.} \end{aligned}$$

Example 17

Factorise $3x - 2dy + 3y - 2dx$.

The terms $3x$ and $3y$ both have 3 in common.
The terms $2dx$ and $2dy$ both have $2d$ in common. Group the given terms in this order.

$$\begin{aligned} 3x - 2dy + 3y - 2dx &= 3x + 3y - 2dx - 2dy \\ &= 3(x + y) - 2d(x + y) \\ &= (x + y)(3 - 2d) \end{aligned}$$

Exercise 4h

Factorise the following by grouping terms in pairs.

- ① $3x + 9b + 5ax + 15ab$
- ② $2ce - 2cf + de - df$
- ③ $4u + 4v + vx + ux$
- ④ $mn - 3my - 3nx + 9xy$
- ⑤ $6a^2 - 3a + 4a - 2$
- ⑥ $cd - ce - d^2 + de$
- ⑦ $12eg - 4eh - 6fg + 2fh$
- ⑧ $4ab + 6bn - 2a - 3n$
- ⑨ $3ax - 2a - 6bx + 4b$
- ⑩ $ax - a + x - 1$
- ⑪ $ac + ad - bc - bd$
- ⑫ $4a - 7b + 28bx - 16ax$
- ⑬ $p + q + 5ap + 5aq$
- ⑭ $2c^2 - 8cm - 3cm + 12m^2$
- ⑮ $x^2 + 2x - 5x - 10$
- ⑯ $2pr - sq + 2qr - ps$
- ⑰ $y^2 - 5y + 4y - 20$
- ⑱ $6ac + bd - 3bc - 2ad$
- ⑲ $2ab - 10cd - 5bc + 4ad$
- ⑳ $6xy - 2z - 4y + 3xz$

[WAEC]

[WAEC]

4.5 Substitution

Example 18

Find the value of $4(3d - e) - 2f$ when $d = 2$, $e = 4$ and $f = 3$.

Either

$$\begin{aligned} 4(3d - e) - 2f &= 4(3 \times 2 - 4) - 2 \times 3 \\ &= 4(6 - 4) - 6 \\ &= 4 \times 2 - 6 \\ &= 8 - 6 \\ &= 2 \end{aligned}$$

or

$$\begin{aligned} 4(3d - e) - 2f &= 12d - 4e - 2f \\ &= 12 \times 2 - 4 \times 4 - 2 \times 3 \\ &= 24 - 16 - 6 \\ &= 2 \end{aligned}$$